

Lumine Reader

Structured Extraction Stress Test

This synthetic document deliberately contains structures and artifacts that Lumine Reader must preserve, clean, validate, or transform before text-to-speech.

CHAPTER 1

Why Structure Matters

A book is not merely a stream of characters. A useful extraction pipeline should retain titles, headings, paragraphs, lists, tables, figures, and reading order.

1.1 Wrapped and Hyphenated Text

Document extraction can introduce awkward line wrapping. A word such as informa-tion may be split across visual lines even though it should become information. The cleaner must avoid destroying genuine hyphens such as state-of-the-art.

1.2 Links, Citations, and Written-Only Material

Read the Lumine documentation or visit <https://example.com/very/long/path?chapter=1>. Revenue increased by 12.5% [12]. The project received £2.4 million, \$50, and €30. R&D continues.

A narrator should not blindly pronounce every URL, citation marker, or formatting artifact.

1.3 Lists and Reading Order

- Preserve semantic content.
- Preserve heading hierarchy.
- Remove speech-irrelevant artifacts only at the appropriate stage.
- Do not confuse document cleaning with TTS normalization.

CHAPTER 2

Tables, Figures, and Symbols

The following table should be recognized as structured content rather than merged arbitrarily into surrounding prose.

Stage	Input	Output
Extract	PDF / EPUB / DOCX	Structured document
Clean	Extracted content	Canonical content
Validate	Canonical content	PASS / WARNING / FAIL
TTS Normalize	Canonical content	Speech-ready text

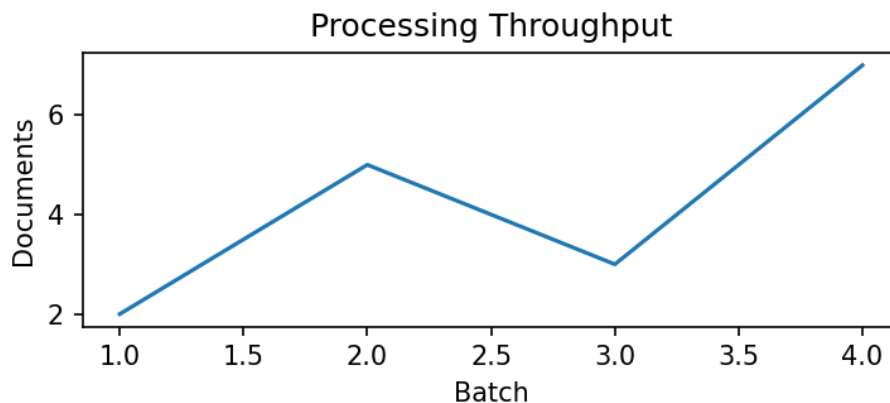


Figure 1. Example processing throughput. The system should preserve the caption and distinguish it from ordinary body text.

2.1 Numbers and Mathematical Content

Version 2.0 processed 1,250 documents. Accuracy was 98.7%. A simple equation is $E = mc^2$. Another expression is $x^2 + y^2 = z^2$. The MVP should preserve equations rather than invent a potentially incorrect pronunciation.

2.2 Code-Like Content

A written document may contain code such as: `normalize_for_tts(text)`. It may also contain file paths like `src/tts/normalize.py` and identifiers such as `MIN_RETENTION_RATIO`. These should remain distinguishable from ordinary prose.

CHAPTER 3

Extraction Noise and Edge Cases

Smart quotes should remain meaningful: “Lumine Reader” and ‘structured extraction’. Unicode characters include café, naïve, résumé, em dash — and ellipsis

HTML-like source artifacts might look like `<strong>important</strong>`, while Markdown-like text might appear as `**important**`, `# Heading`, `[reference](https://example.com)`, or `![Architecture diagram](architecture.png)`.

A malformed extraction can contain a replacement character: ■. This should be detectable by validation rather than silently propagated.

3.1 Repeated Header Simulation

LUMINE READER — STRUCTURED EXTRACTION TEST

Repeated headers and standalone page numbers are common PDF artifacts. Docling should help us understand which of these can be identified structurally instead of relying entirely on regular expressions.

End of synthetic test document.